

Philippines Office Internet — why classes break, and how we fix it

Prepared for **Karl** (PH Manager). Our server logs now measure every class, and they point clearly at the office upload line. This plan lists what we already fixed on our side, what we need you to measure this week, and what we should buy if the measurement confirms the cause.

Evidence: server line-quality logs (1 row per minute per person) · admin login IP records · deployment logs. Where something is a conclusion rather than a measurement, it is marked **[inference]**.

1 What happened (measured)

Class on 3 Sept, 14:00 KST — Teacher Farrah with Korean student **ysyt01**, 21 minutes. The student saw a blurry, freezing picture; Farrah's video kept switching itself off and on.

What we measure	Normal class (same days)	This class	Meaning
Round-trip delay, average	100–150 ms	609 ms (teacher) / 588 ms (student)	4–6× slower than normal
Round-trip delay, worst minute	under 200 ms	1,091 / 1,267 ms	Over a second of lag
Audio concealed (sound lost and patched by the browser)	0.3–1.6 %	30–44 %	Roughly 1 word in 3 damaged
Video freezes per minute	about 1	6–21	Picture stalls every few seconds
Automatic “audio-only” switches	0	6	Camera turning itself off and on

Your own test at 15:00 the same day (Karl ↔ Farrah, both in the Philippines) showed round-trip delay of **60 ms → 1,676 ms → 60 ms within three minutes**. Both of you were inside the Philippines, so the Korea–Philippines undersea route is not the problem. The problem is on the Philippine side.

2 Which line, exactly

Our login records show that **about ten teacher accounts plus you and Melca all sign in from a single public IP address** (216.247.55.144; it was 216.247.53.158 before 28 Aug). That means the teaching PCs share **one office internet line**.

Important — the line is not always bad. On 1 Sept, Hannah, Win and Krystel taught on that same line with 170–190 ms delay and 1–7 % audio loss, which is fine. And on 3 Sept at 14:00 our records show only **one** Mangoi class running, so it was not “too many classes at once”.

Teacher (last 4 days, class rooms)	Avg delay	Audio concealed	Auto audio-only
Farrah	609 ms	33.6 %	6
Teacher C (China, separate line)	384 ms	1.3 %	0
Teacher (mangoi_153)	206 ms	12.9 %	1
Hannah	193 ms	6.3 %	0
Win	174 ms	0.8 %	1
Krystel	172 ms	7.4 %	1
Teacher (mangoi_114)	102 ms	0.4 %	0

Farrah's line record currently covers only this one class (teacher-side logging started 1 Sept), so we cannot yet say whether her situation is permanent or was specific to that hour. That is exactly what step 4 settles.

3 Most likely cause [inference]

The office upload capacity is being filled up, so packets queue and delay explodes. Engineers call this “bufferbloat”. The signature is exactly what we measured: delay jumping from 0.06 s to 1.6 s while packet loss stays comparatively low.

A video class has to **upload** the teacher's camera continuously. Home and small-office plans in the Philippines usually have a much smaller upload than download. Once upload is full, everything on that line — including the class — starts queueing.

Three candidates, and we cannot tell them apart from Korea:

1. The contracted **upload speed is simply too small** for the number of simultaneous classes.
2. Something else on the same line is uploading — **cloud backup, Windows Update, file sync, YouTube upload, another video call**.
3. The **ISP itself is congested** in that time slot.

Ruled out on our side (measured): our relay servers were healthy; the two people were in the same room; there were no ghost connections; the teacher PC no longer records video (we removed that load on 3 Sept morning); and our video quality setting is already on “Low”, so we are not sending a heavy stream.

4 What we need from you this week

These cost nothing and they decide where the money should go. Please do them and send the numbers back.

#	Task	How
1	Measure the line during class hours	On a teaching PC open fast.com , click Show more info , and screenshot Upload and Loaded latency . Do this at 14:00, 16:00 and 20:00 for three days. Loaded latency above 500 ms confirms the cause above.
2	Confirm how each teaching PC connects	Wired LAN cable or Wi-Fi? If Wi-Fi, how far from the router, and how many walls?
3	List what else uses the line	Number of PCs, phones, CCTV; any cloud backup / Google Drive / OneDrive / Windows Update running in the daytime.
4	Send the internet contract details	ISP name, plan name, and the contracted upload speed (not just download). Ask the ISP if it is not on the bill.
5	Check the class app setting	In the class screen settings, Video quality should be Low .

Please fill this in and send it back:

Item	Answer
ISP / plan name	_____
Contracted download / upload	_____ Mbps / _____ Mbps
fast.com upload during class	_____ Mbps
fast.com loaded latency during class	_____ ms
Teaching PCs: wired / Wi-Fi	_____ wired, _____ Wi-Fi
Max classes running at the same time	_____
Worst time of day for quality	_____

5 Fixes, cheapest first

#	Action	Owner	Effect	Cost
1	Wired LAN for every teaching PC. Wi-Fi can halve usable upload through one wall.	Karl	High	A few cables
2	Stop other uploads during class hours — pause cloud backup and Windows Update, no file sync or personal video calls on that line.	Karl	High	None
3	Enable QoS on the router so the teaching PCs get priority over everything else.	Karl	High	None (existing router)
4	Upgrade to a symmetric fibre plan with real upload capacity. Rule of thumb: about 2 Mbps of upload per simultaneous class, plus headroom.	HQ approves	High	Monthly difference — quote needed
5	Add a second ISP + dual-WAN router. When one carrier congests, classes ride the other. This is the closest thing to a guarantee.	HQ approves	High	Second line + router, one-off
6	Backup mobile data (LTE/5G hotspot) for emergencies, so a class is never cancelled outright.	HQ approves	Medium	Small monthly

Recommended order: 1–3 immediately (this week, essentially free), then decide 4 or 5 using the numbers from section 4. Do not buy a bigger plan before measuring — if the real cause is a background upload or Wi-Fi, a bigger plan changes nothing.

6 What we already did, and what we are building

Done (deploying tonight): the video no longer flickers on and off. Previously the app restored video after only 8 seconds of calm and then dropped it again; now it waits for **32 seconds of stable line** before turning the camera back on. In a simulation of exactly your 3 Sept conditions this reduced the switching from **12 times to 1** over four minutes.

Also shipped this week: video quality now recovers on high-latency lines instead of staying blurry; the teacher PC no longer records the class (that load moved off her machine); and a badge now tells everyone when the picture is low quality, so nobody thinks the app is broken.

Building next: we are adding **per-line quality tracking**. Today our dashboard shows quality per person; it will also group by internet line, so we can see statements like “the office line degrades every day between 14:00 and 16:00”. That gives you a documented, time-stamped record to take to the ISP — which is much stronger than a phone complaint.

To be clear: none of our software changes make the line itself faster. They stop the symptoms from being worse than the line. Sections 4 and 5 are the real fix.

7 How we will know it worked

- In the admin dashboard, **Teacher Line Quality**: Farrah's row should show average delay **below 300 ms** and audio concealed **below 5 %**.
- Compare the days before and after the LAN cables — the difference should be visible immediately.
- The camera should stop switching itself off and on after tonight's deployment. If it still does, tell us: that would mean a different cause.

Target date: steps 1–3 and the measurements by **Monday 7 September**, decision on step 4/5 the same week.

Measured from: class quality logs (3 Sept, 40 one-minute rows) · Karl–Farrah test (3 Sept 15:00, 6 rows) · teacher comparison (4 days, 8 teachers) · admin login IP records (14 days) · deployment runs 2989 / 2995. Anything not directly measured is marked [inference]. Questions: reply on KakaoTalk.